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Fig. 5: Various types of CAT standards (Image courtesy: Fluke)

CAT-III is necessary. IEC 1010 standard (International Electro-Technical Commission) is another standard for low-voltage T&M equipment. Also, make sure your multimeter has a CE marking (a certification which means, the product meets European standards).

Goliya says, "The user must look for safety features, such as overload protection in various ranges, safety categories like CAT IV 600 V or CAT III 1000V, surge protection (8kV-12kV) as well as electromagnetic compatibility (EMC) protection." He adds, "Shielding from strong magnetic fields to give accurate readings even in strong magnetic field areas and fire-resistant casings for operator safety are also very important in a multimeter. The tendency of most purchasers is to buy the cheapest multimeter, thereby compromising on safety."

Kumar informs that for any 600V digital multimeter, there has to be a CAT III protection level, and every 1000V multimeter should have a CAT IV protection level.

Use areas. It comes down to one question: What would you mostly be using the digital multimeter for?

The major factor to choose a multimeter should be driven by your application. "It is a crucial parameter to consider. Some people go for a multi-range multimeter even though they might not require one. If a user needs a little bit of everything, then he or she should go for a general-purpose multimeter," says Kwatra. If you are a professional electronics design engineer, then you should consider a multimeter from reputed brands that provide measurement convenience, safety and great build quality, even though these might be slightly expensive.

For a hobbyist tinkering with circuits, where accuracy is not of great significance, low-cost multimeters would suffice, although most of these do not adhere to safety standards. As you transition to high-end multimeters, these become very application-specific. There are a variety of application-specific multimeters available for areas such as industrial, automotive, audio and sound, among others. Therefore it is extremely important for a user to know the purpose and usage areas before buying a multimeter.

Auto-range multimeters. Most of the multimeters available today are auto-range as these are simpler to use and avoid wrong range selection. Manual-range multimeters, which is a pretty old technology, require you to select the range manually each time

you make different measurements, which can get tiresome in the long run. An auto-range multimeter with the manual-range override control can be very handy as it provides you with an option of to fix the range manually.

Other specifications. Accuracy of a digital multimeter is a key parameter. Apart from accuracy, some other features to look out for include the range of measurement, number of digits in the display, safety ratings, rugged body, long battery backup, remote connectivity and better IP (international protection or ingress protection) rating. Dimension and weight could be important for engineers who prefer portability with multimeters. "A user, having fixed his or her requirements, should select the instrument not only for present requirements but also future needs," informs Goliya.

A reasonably large, clear and well-lit display for comfortable readings with availability of warranty, service and affordable spare parts could prove to be beneficial in the long run.

Choose wisely, measure well

Multimeters are one of the most widely used T&M equipment. Although there are a variety of brands and multimeters out there, the area where your digital multimeter will be used remains the cornerstone of your purchase. Low-end multimeters come in handy for DIY electronics. But most of the cheap multimeters do not conform to safety standards, therefore it would be preferable to select one from a known brand. ●